



City Metro Transit System - SQL Project

The **City Metro Transit System** is a SQL database project developed using **MySQL**. It simulates the operations of a city's metro railway system by managing stations, trains, passengers, and ticket bookings. The project demonstrates both basic and advanced SQL concepts through a real-world transportation scenario.

Objectives

- Design a relational database for a metro transit system.
- Store and manage station, train, passenger, and booking information.
- Maintain data integrity using SQL constraints.
- Analyze booking data using SQL queries.
- Practice advanced SQL concepts such as CTEs, Views, Window Functions, Transactions, and Indexes.

Technologies Used

- MySQL 8.0
- MySQL Workbench
- SQL

Database Structure

The project consists of four interconnected tables:

- **Stations** – Stores metro station details.
- **Trains** – Stores train information.
- **Passengers** – Stores passenger details.
- **TicketBookings** – Stores ticket booking records.

Dataset Summary

Table	Records
Stations	25
Trains	20
Passengers	465
TicketBookings	1490

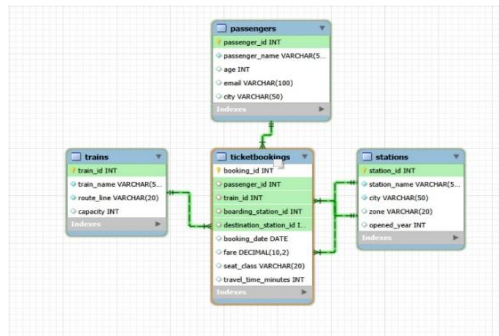
Total Records: Approximately **2,000**

SQL Concepts Covered

- Primary Key & Foreign Key
- Unique, Check & Default Constraints
- Aggregate Functions
- String Functions
- Date Functions
- Joins
- Group By & Having

- Subqueries
- Common Table Expressions (CTEs)
- Views
- Window Functions
- Transactions (Commit & Rollback)
- Indexing

Database Tables



Project Details

1.Title :-Retrieving Passenger and Train Details Using JOIN

Description

This query combines data from the **TicketBookings**, **Passengers**, and **Trains** tables using JOIN. It displays the booking ID, passenger name, and train name for each booking.

Purpose

It demonstrates how SQL joins retrieve related information from multiple tables.

Booking_id	Passenger_name	Train_name
1049	Sunita Sharma	Blue Line Express
1066	Lakshmi Reddy	Blue Line Express
1083	Naveen Nayar	Blue Line Express
1094	Swati Kapoor	Blue Line Express
1106	Ganesh Bhatt	Blue Line Express
1117	Ramesh Iyer	Blue Line Express
1130	Sneha Bhatt	Blue Line Express
1132	Deepak Joshi	Blue Line Express
1144	Shalini Rao	Blue Line Express

2. Title :-Total Fare Collected at Each Boarding Station

Description

This query uses the SUM() aggregate function along with GROUP BY to calculate the total fare collected at each boarding station.

Purpose

It helps analyze revenue generated from different metro stations.

Station_name	Total_fare
Airport Link	2149.87
Botanical Garden	1907.41
Central Junction	1361.12
City Center	1593.21
Convention Center	1105.18
East Gate	2141.39
Greenfield	1140.51
Harbor Point	2802.43
Heritage Circle	1633.3

3.Title :-Passenger Booking Summary Using CTE

Description

This query uses a Common Table Expression (CTE) to calculate the total number of bookings made by each passenger and displays the passenger name along with the booking count.

Purpose

It demonstrates how CTEs simplify complex queries and improve readability

Passenger_name	Age	Email	Total_Bookings
Nisha Singh	25	nisha.singh1@metromail.com	2
Divya Sharma	22	divya.sharma2@metromail.com	1
Arjun Das	16	arjun.das3@metromail.com	1
Ashok Kapoor	67	ashok.kapoor4@metromail.com	1
Sunita Kumar	22	sunita.kumar5@metromail.com	2
Swati Iyer	45	swati.iyer6@metromail.com	4
Arjun Choudhary	8	arjun.choudhary7@metromail.com	4
Mahesh Nair	72	mahesh.nair8@metromail.com	2
Ritu Bose	33	ritu.bose9@metromail.com	1

4.Title :-Ranking Passenger Bookings by Fare

Description

This query uses the DENSE_RANK() window function to rank each passenger's bookings based on fare without leaving gaps in the ranking sequence.

Purpose

It demonstrates the use of window functions for analytical reporting.

Booking_id	Passenger_id	Fare	Fare_rank
1081	1	154.77	1
1191	1	39.32	2
1251	2	174.96	1
1305	3	171.28	1
1062	4	145.64	1
1464	5	126.1	1
1124	5	38.54	2
1367	6	162.38	1
1279	6	156.04	2

5. Title :-Creating a Composite Index

Description

This screenshot shows the creation of a composite index on the `boarding_station_id` and `destination_station_id` columns. The index helps improve the performance of queries that filter on these columns.

Purpose

It demonstrates query optimization using indexing.

Table	Non_unique	Key_name	Seq_in_index	Column_name	Collation	Cardinality	Sub_part	Packed	Null	Index_type	Comment	idx_comme	Visible	Expression
ticketbookings	0	PRIMARY	1	booking_id	A	490	NULL	NULL		BTREE	N/A	N/A	YES	NULL
ticketbookings	1	train_id	1	train_id	A	20	NULL	NULL	YES	BTREE	N/A	N/A	YES	NULL
ticketbookings	1	destination_station_id	1	destination_station_id	A	25	NULL	NULL	YES	BTREE	N/A	N/A	YES	NULL
ticketbookings	1	idx_passenger_id	1	passenger_id	A	302	NULL	NULL	YES	BTREE	N/A	N/A	YES	NULL
ticketbookings	1	idx_boarding_destination	1	boarding_station_id	A	25	NULL	NULL	YES	BTREE	N/A	N/A	YES	NULL
ticketbookings	1	idx_boarding_destination	2	destination_station_id	A	328	NULL	NULL	YES	BTREE	N/A	N/A	YES	NULL



Learning Outcomes

Through this project, I learned how to:

- Design and implement a relational database.
- Build relationships using primary and foreign keys.
- Write SQL queries for data retrieval and analysis.
- Use advanced SQL concepts to solve real-world problems.
- Optimize database performance using indexes.



Conclusion

This project helped me strengthen my SQL and database management skills by applying both fundamental and advanced SQL concepts to a real-world metro transit system. It provided practical experience in designing, managing, and analyzing relational databases using MySQL.